

21. Silver nitrate produces a black stain on skin due to

- (a) Being a strong reducing agent
- (b) Its corrosive action
- (c) Formation of complex compound
- (d) Its reduction to metallic silver

Answer: D — (d) Its reduction to metallic silver

Chemical reason: AgNO_3 darkens skin because light and organic matter reduce Ag^+ to finely divided metallic silver, which appears black or dark brown.

22. When hypo solution is added to cupric sulphate solution, the blue colour of the latter is discharged, due to formation of

- (a) CuS_2O_3
- (b) $\text{Na}_2\text{S}_4\text{O}_6$
- (c) NaCuS_2O_3
- (d) Cu_2O

Answer: C — (c) NaCuS_2O_3

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in NaCuS_2O_3 has the characteristic oxidation state/ligand field that produces the observed colour.

23. Metal oxides which decomposes on heating is

- (a) ZnO
- (b) Al_2O_3
- (c) CuO
- (d) Na_2O
- (e) HgO

Answer: E — (e) HgO

Chemical reason: HgO is thermally unstable and decomposes on strong heating to mercury and oxygen: $2\text{HgO} \rightarrow 2\text{Hg} + \text{O}_2$.

24. Anhydrous sample of ferric chloride is prepared by heating

- (a) $\text{Fe} + \text{HCl}$
- (b) $\text{Fe} + \text{Cl}_2$
- (c) $\text{FeCl}_2 + \text{Cl}_2$
- (d) Hydrated ferric chloride

Answer: B — (b) $\text{Fe} + \text{Cl}_2$

Chemical reason: Anhydrous FeCl_3 cannot be prepared simply by heating the hydrate because hydrolysis occurs. It is prepared by direct chlorination of iron under dry conditions: $2\text{Fe} + 3\text{Cl}_2 \rightarrow 2\text{FeCl}_3$.

25. Light green crystals of ferrous sulphate lose water molecule and turn brown on exposure to air. This is due to its oxidation to

- (a) Fe_2O_3
- (b) $\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$
- (c) $\text{Fe}(\text{OH})\text{SO}_4$
- (d) $\text{Fe}_2\text{O}_3 + \text{FeO}$

Answer: C — (c) $\text{Fe}(\text{OH})\text{SO}_4$

Chemical reason: Ferrous sulfate crystals are pale green because of Fe^{2+} . On exposure to moist



air, Fe^{2+} is oxidized to Fe^{3+} and basic ferric species such as $\text{Fe}(\text{OH})\text{SO}_4$ form, giving a brown colour.

26. In alkaline condition KMnO_4 , reacts as follows : $2\text{KMnO}_4 + 2\text{KOH} \rightarrow 2\text{K}_2\text{MnO}_4 + \text{H}_2\text{O} + \text{O}$ Therefore its equivalent weight will be

- (a) 31.5
- (b) 52.7
- (c) 72.0
- (d) 158.0

Answer: D — (d) 158.0

Chemical reason: In strongly alkaline medium permanganate is reduced from Mn(VII) to manganate, Mn(VI), so the change is 1 electron per Mn. Therefore $n = 1$ and the equivalent weight equals the molar mass.

27. Equivalent weight of KMnO_4 acting as an oxidant in acidic medium is equal to

- (a) Molecular weight of KMnO_4
- (b) 12 Molecular weight of KMnO_4
- (c) 13 Molecular weight of KMnO_4
- (d) 15 Molecular weight of KMnO_4

Answer: D — (d) $(1)/(5)$ Molecular weight of KMnO_4

Chemical reason: Equivalent weight is calculated as molar mass divided by the number of electrons exchanged per formula unit in the stated redox reaction.

28. In which of the following ionic radii of chromium would be smallest

- (a) K_2CrO_4
- (b) CrO_2
- (c) CrCl_3
- (d) CrF_2

Answer: A — (a) K_2CrO_4

Chemical reason: The correct choice is K_2CrO_4 . It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

29. $\text{CoO} \cdot \text{Al}_2\text{O}_3$ is called

- (a) Cobalt aluminate
- (b) Thenard's blue
- (c) Both
- (a) and
- (b)
- (d) None of these

Answer: C — (c) Both (a) and (b)

Chemical reason: $\text{CoO} \cdot \text{Al}_2\text{O}_3$ is cobalt aluminate, a spinel pigment known as Thenard's blue. Therefore both names refer to the same compound.



**30. $\text{ZnO} \cdot \text{CoO}$ is called**

- (a) Cobalt zincate
- (b) Rinman's green
- (c) Both
- (a) and
- (b)
- (d) None of these

Answer: C — (c) Both (a) and (b)

Chemical reason: $\text{ZnO} \cdot \text{CoO}$ is the green cobalt-zinc oxide pigment known as cobalt zincate or Rinmann's green. Thus both names are correct.

31. $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ is called

- (a) Mohr's salt
- (b) Green salt
- (c) Alum
- (d) Glauber's salt

Answer: A — (a) Mohr's salt

Chemical reason: $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ is ferrous ammonium sulfate, commonly called Mohr's salt. It is a stable double salt containing Fe^{2+} and NH_4^+ .

32. Molybdenum compounds are used in

- (a) Dye industry
- (b) For colouring leather
- (c) For colouring rubber
- (d) All of these

Answer: D — (d) All of these

Chemical reason: The application follows from the characteristic chemical property of the material—such as catalytic activity, stability, conductivity, or corrosion resistance. That

property is specifically associated with All of these.

33. When copper turnings and concentrated HCl is heated with copper sulphate the compound formed is

- (a) Cupric chloride
- (b) Cuprous chloride
- (c) Copper sulphate
- (d) SO_2

Answer: B — (b) Cuprous chloride

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives Cuprous chloride. The choice is consistent with the expected oxidation state and stability of the product.

34. The compound of copper which turns green on keeping in air is

- (a) Copper sulphate
- (b) Copper nitrate
- (c) Cupric chloride
- (d) Cuprous chloride

Answer: D — (d) Cuprous chloride

Chemical reason: The correct choice is Cuprous chloride. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.



35. Cu_2Cl_2 with HCl in presence of oxidising agents gives

- (a) CuCl_2
- (b) H_2CuCl_2
- (c) Hydrogen gas
- (d) Chlorine gas

Answer: A — (a) CuCl_2

Chemical reason: The result is obtained by assigning oxidation numbers before and after the reaction and balancing electron transfer. The species in CuCl_2 has the oxidation-state change required by the stated redox conditions, so that choice is correct.

36. $\text{K}_2\text{Cr}_2\text{O}_7$ on heating with aqueous NaOH gives

- (a) CrO_4^{2-}
- (b) $\text{Cr}(\text{OH})_3$
- (c) $\text{Cr}_2\text{O}_7^{2-}$
- (d) $\text{Cr}(\text{OH})_2$

Answer: A — (a) CrO_4^{2-}

Chemical reason: Amphoteric metals/oxides react with both acids and strong bases, often forming soluble hydroxo or oxo complexes in alkali. The species/result CrO_4^{2-} is consistent with that behaviour.

37. KMnO_4 reacts with oxalic acid according to the equation : $2\text{MnO}_4^- + 5\text{C}_2\text{O}_4^{2-} + 16\text{H}^+ \rightarrow$

$2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O}$ Here 20 ml of 0.1 M KMnO_4 is equivalent to

- (a) 20ml of 0.5 M $\text{C}_2\text{H}_2\text{O}_4$
- (b) 50ml of 0.1 M $\text{C}_2\text{H}_2\text{O}_4$
- (c) 50ml of 0.5 M $\text{C}_2\text{H}_2\text{O}_4$
- (d) 20ml of 0.1 M $\text{C}_2\text{H}_2\text{O}_4$

Answer: B — (b) 50 mL of 0.1 M $\text{C}_2\text{H}_2\text{O}_4$

Chemical reason: The balanced reaction shows 2 mol MnO_4^- react with 5 mol oxalate/oxalic acid. Thus 20 mL of 0.1 M KMnO_4 contains 0.002 mol and requires 0.005 mol oxalic acid, which is 50 mL of a 0.1 M solution.

38. The equivalent weight of potassium permanganate for acid solution is

- (a) 158
- (b) 31.6
- (c) 52.16
- (d) 79

Answer: B — (b) 31.6

Chemical reason: For permanganate in acidic medium, Mn(VII) is reduced to Mn^{2+} , a 5-electron change. Therefore $n = 5$ and equivalent weight = molar mass/5.



**39. Which statement is not correct**

- (a) Potassium permanganate is a powerful oxidising substance
- (b) Potassium permanganate is a weaker oxidising substance than potassium dichromate
- (c) Potassium permanganate is a stronger oxidising substance than potassium dichromate
- (d) Potassium dichromate oxidises a secondary alcohol into a ketone

Answer: B — (b) Potassium permanganate is a weaker oxidising substance than potassium dichromate

Chemical reason: The correct choice is Potassium permanganate is a weaker oxidising substance than potassium dichromate. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

40. The formula of corrosive sublimate is

- (a) HgCl_2
- (b) Hg_2Cl_2
- (c) Hg_2O
- (d) Hg

Answer: A — (a) HgCl_2

Chemical reason: Corrosive sublimate is the historical name for mercuric chloride, HgCl_2 , a volatile and highly toxic mercury(II) salt.

